
Unit – 2 : Session – 11 : SLO – 2 | Hardware-Based Synchronization — Fill in the Blanks

Synchronization Hardware

1. Hardware solutions for synchronization aim to achieve mutual exclusion without the need for complex software mechanisms.
2. The Test-and-Set instruction is used to test a lock variable and set it if it is free, in one atomic operation.
3. In busy waiting (spinning), a process continuously checks a condition without releasing the CPU.
4. Mutual exclusion ensures that only one process is allowed to enter the critical section at a time.
5. The Compare-and-Swap instruction compares the value of a memory location with a given value and modifies it if they match.
6. A spinlock is a simple lock mechanism where the CPU repeatedly checks a flag before proceeding.
7. The simplest form of hardware synchronization is to disable interrupts so that only one process executes at a time.
8. A lock variable is typically a shared variable used to implement locking between threads.
9. Hardware instructions like test-and-set must be atomic to prevent race conditions.
10. Hardware mechanisms are often faster but harder to manage than software-based approaches.